Industrial Smart Display

AIO-ESP32P4-101

* For illustration purposes only. The user interface shown is a visual mockup and is not pre-installed or included with the product. Actual software and interface may vary.

PN: CS12800-ESP32P4-101A

Content can change at anytime, check our website for latest information of this product.
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Contents

AIO-ESP32P4-101

Front View

* For illustration purposes only. The user interface shown is a visual mockup and is not pre-installed or included with the product. Actual software and interface may vary.

Rear View


Chipsee
PN:CS12800-ESP32P4-101A
SN: 2026082100000002
https://chipsee.com
Origin:Made in China

Side View 1

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Side View 2

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Product Overview

The AIO-ESP32P4-101 Smart Display (PN: CS12800-ESP32P4-101A) is an all-in-one programmable capacitive touch smart display featuring an ESP32 P4 processor.

The product features a lightweight plastic enclosure available in black or white. It can be used as a portable device or mounted in a fixed location.

It features a 10.1” IPS display with a maximum brightness of 400 cd/m^2. The touch screen is a responsive capacitive screen and supports multi-touch.

The product is powered by an ESP32-P4 processor, making it suitable for applications such as:

- Smart Home
- Industrial Automation
- Healthcare
- Consumer Electronics
- Smart Agriculture
- Retail Self-Service Terminals (POS, Vending Machines)
- Service Robots
- Multimedia Player
- Cameras for Video Streaming
- High-Speed USB Host and Device
- Smart Voice Interaction Terminal
- Edge Vision AI Processor
- HMI Control Panel

The product can be programmed with ESP-IDF, ESPHome and other ESP32 ecosystem tools and frameworks that support the ESP32-P4.

Ordering Options

Chipsee products can be customized during the ordering process. The product will be shipped with the pre-installed factory defaults if no extra requirements are specified. The table in the Hardware Features section provides information about the default options bundled with the product.

> **Note**
>
> You can order AIO-ESP32P4-101 from the official Chipsee Store or from your nearest distributor.

Hardware Features

The AIO-ESP32P4-101 Smart Display offers a broad range of performance and connectivity features for flexible integration. Some of the key features are listed below.

AIO-ESP32P4-101	
CPU	Dual-core 32-bit RISC-V, up to 400 MHz; single-core 32-bit LP RISC-V, up to 40 MHz
Storage	32MB external SPI Flash
RAM	768 KB HP L2MEM + 32 KB LP SRAM + 8 KB SPM; 32MB in-package PSRAM
Display	10.1” IPS LCD, 1280 x 800 px, brightness 400 cd/m^2
Touch	10-point capacitive touch with 1.0mm armored glass
LAN	1 x 100Mbps LAN
WiFi/BT	WiFi 6, Bluetooth LE 5 **(optional, ESP32-C6 is not mounted by default)**
4G/LTE	Internal Cat 1 4G/LTE module, **not mounted by default**
Audio	3.5mm audio out interface; internal dual channel speaker; front microphone
Buzzer	Onboard buzzer, driven by GPIO
Camera	1 x MIPI-CSI (2-lane) interface for connecting a MIPI-CSI (2-lane) 15-pin 1.0mm camera such as a Raspberry Pi camera **(optional)**
GPIO	0 x GPIO **(default)**, 8 x GPIO **(optional,** by configuring 2 x RS232, 1 x RS485 and 1 x Relay to 8 x GPIO)
Relay	1 x relay with “Normally Closed” and “Normally Open” output; relay NC and NO can be configured to 1 x CAN FD **(optional)**; relay NC and NO can be configured to 2 x GPIO **(optional)**
RS232	2 x RS232 **(default)**; RS232_2 can be configured to RS485 **(optional)**; 2 x RS232 can be configured to 4 x GPIO **(optional)**
RS485	1 x RS485 **(default)**; up to 2 x RS485 **(optional,** by configuring RS232_2 to RS485); 1 x RS485 can be configured to 2 x GPIO **(optional)**
CAN	0 x CAN **(default)**; 1 x CAN FD bus, arbitration bit rate up to 1Mbps, data bit rate up to 8Mbps, by configuring 1 x relay to CAN **(optional)**
USB	2 x USB 2.0 Type-A high-speed host ports; 1 × USB 1.1 Type-C port for firmware flashing and debugging; does not support powering the device.
Micro SD card socket	1 x TF (micro SD) card slot
RTC	High-accuracy RTC with supercapacitor, can work 1 week after power off **(default)**. High accuracy RTC with lithium coin battery, can work 3 years after power off **(optional)**.
Power Input	9V to 32V DC
Current	260mA max at 15V, 200mA typical at 15V

AIO-ESP32P4-101	
Power Consumption	4W max, 3W typical
Operating Temp.	From -20°C to +70°C
Dimensions	260.54 x 178.54 x 26.9mm
Weight	620g
Mounting	VESA, Stand
Plastic Case Color	Black / White
Certification	CE, RoHS

Table 503 Key Features

Optional Features

- WiFi/Bluetooth: If Wi-Fi/Bluetooth connectivity is required, an ESP32-C6 module can be installed at the factory. The ESP32-C6 is not populated by default.
- Camera: The product provides a 2-lane MIPI-CSI interface for compatible 15-pin camera modules such as Raspberry Pi cameras. The optional camera-related hardware is not populated by default. Please specify this requirement when placing your order.
- 4G/LTE: 4G/LTE (Cat 1) module is supported on this product, it's not mounted by default, if you need the interface or the 4G/LTE module you can contact us when you place an order.

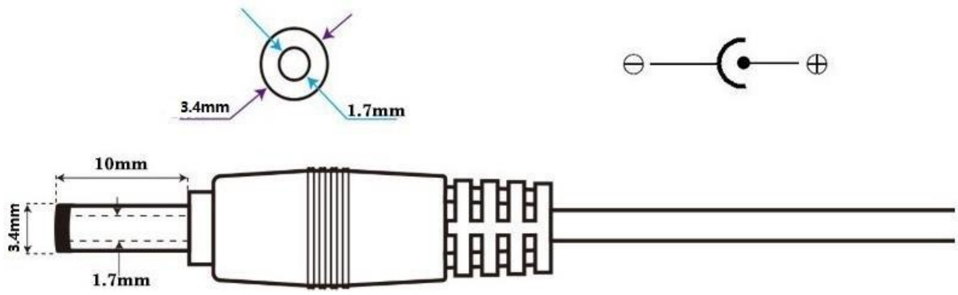
> **Warning**
>
> Installation, repair, and maintenance tasks should be performed by trained personnel only. Chipsee does not bear any responsibility for damage caused by inadequate handling of the product.

Power Input

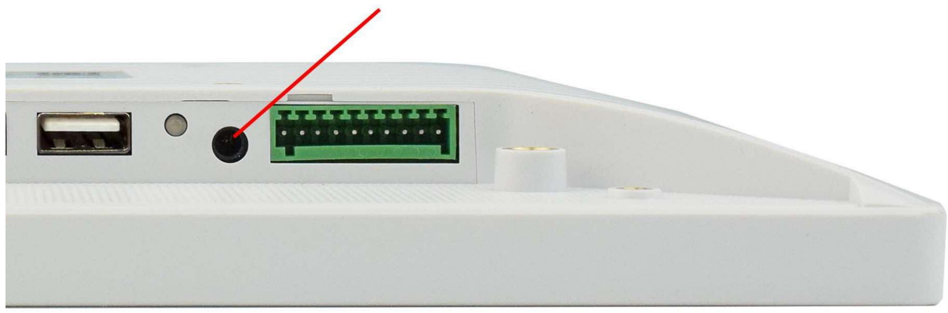
The AIO-ESP32P4-101 Smart Display can be powered by an input voltage of 9V to 32V DC.

The total power consumption is about 4W max, 3W typical, depending on the load and brightness.

The power input uses a DC connector with dimensions of 3.4 mm O.D. × 1.7 mm I.D. × 9.5 mm. The product also includes a 3.4 × 1.7 mm to 5.5 × 2.1 mm adapter, allowing either connector size to be used with a suitable DC power supply.



Power Adapter

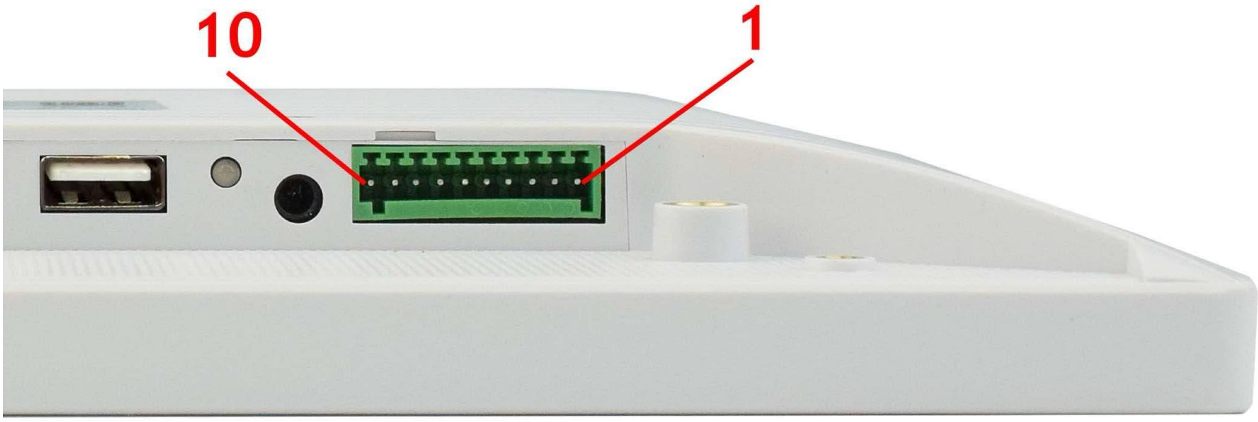


Power Input

Connectivity

There are many connectivity options available on the AIO-ESP32P4-101.

RS232/RS485/Relay



RS232/RS485/Relay Connector

The table below provides a detailed description of each pin and its function.

RS232 / RS485 / Relay Pin Definition:			
Pin Number	**Definition**	**Description**	**ESP32 P4 Pin**
Pin 1	GND	System Ground	GND
Pin 2	RS232_0_RXD	RS232 RXD signal	GPIO38
Pin 3	RS232_0_TXD	RS232 TXD signal	GPIO37
Pin 4	RS232_2_RXD	RS232 RXD signal Can be set as RS485_2+(A)	GPIO2
Pin 5	RS232_2_TXD	RS232 TXD signal Can be set as RS485_2-(B)	GPIO3
Pin 6	RS485_3+	RS485 +(A) signal	GPIO4
Pin 7	RS485_3-	RS485 –(B) signal	GPIO5
Pin 8	Relay NO	Relay Normally Open	
Pin 9	Relay COM	Relay Common Can be set as CAN0 H	GPIO22
Pin 10	Relay NC	Relay Normally Closed Can be set as CAN0 L	GPIO21

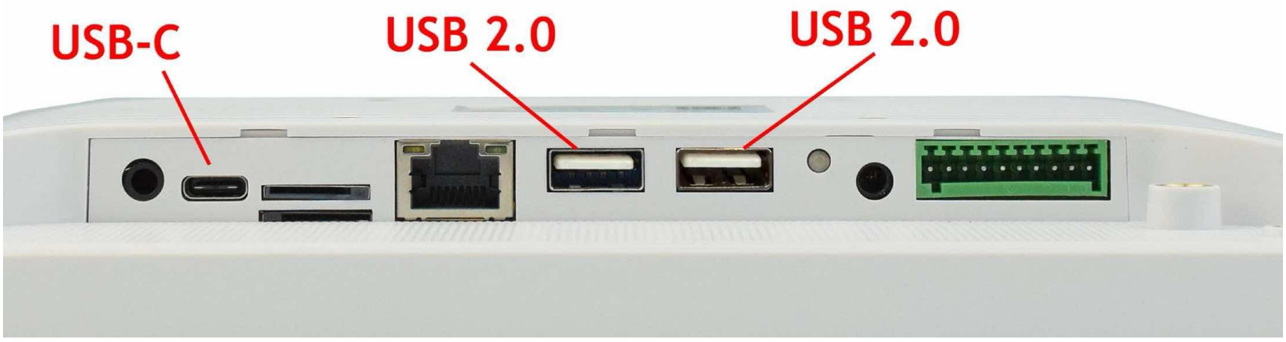
Table 504 RS232/RS485/Relay Connector

Attention

1. The RS232_2 can be set as the RS485 signal (1 x RS232 + 2 x RS485 + 1 x Relay). If you need it to work as RS485, please **Contact us** before shipping.
2. The Relay can be set as the CAN signal (2 x RS232 + 1 x RS485 + 1 x CAN + 0 x Relay). If you need it to work as CAN, please **Contact us** before shipping.
3. RS485_3 automatically controls input/output direction. It does not need software control.
4. The 120 Ω termination resistor for the RS485 is **not mounted** by default.
5. The relay's maximum switching voltage is 125 VAC or 60 VDC. The maximum switching current is 1 A. Rated load is 0.3A at 125VAC and 1A at 30VDC.
6. The RS232, RS485, and relay interfaces can optionally be configured as GPIOs, providing 0, 2, 4, 6, or 8 GPIOs depending on the configuration.

USB

There are 2 x **USB2.0 Type-A HOST connectors** and 1 x **Type-C connector** onboard, as shown in the image below.



USB2.0 Host and Type-C Ports

> **Note**
>
> - USB peripherals such as flash drives, mice, and keyboards can be connected to the USB 2.0 host ports.
> - The USB Type-C port is used for debugging and firmware flashing. It cannot be used to power the product.

LAN

LAN (RJ45) connector provides Ethernet connectivity over standardized Ethernet cable. The integrated Ethernet interface supports up to 100 Mbps data throughput.

RJ45 LAN Connector

4G/LTE

4G/LTE Cat 1 module is **optional**. If you place an order with 4G module included, we will also add a SIM card holder and a 4G/LTE antenna to enable 4G/LTE connectivity.

The SIM card slot does **NOT** support hot-swapping.. **Power off** the device before inserting or removing the SIM card.

The Cat 1 module doesn't support GPS.

SIM Card Direction

> ⚠ **Attention**
>
> The product does not come shipped with the 4G/LTE module by default. If you need to use 4G/LTE, you can contact us when placing an order, we can install the necessary hardware for you.

MicroSD Card

The AIO-ESP32P4-101 Smart Display features 1 x **TF Card (micro SD) slot** (note the direction of the micro SD card). The product does not come shipped with the micro SD card by default.

Micro SD Card Slot

Camera

The product has 1 x MIPI-CSI (2-lane) interface for connecting a MIPI-CSI (2-lane) 15-pin 1.0mm camera such as a Raspberry Pi camera **(optional)**, such as OV5647. When you select a camera, please choose carefully because not all electrically compatible camera modules have mature software support for the ESP32-P4..

Audio

3.5 mm Audio Output

The product features 3.5mm audio connector as shown in the image below.

Audio Out Connector

Internal Speaker

The product also has an internal dual channel 2W speaker, meaning you can play music without an earphone, the sound can be played from the product directly.

Buzzer

The product has an internal buzzer, suitable for playing beep sounds for alarms.

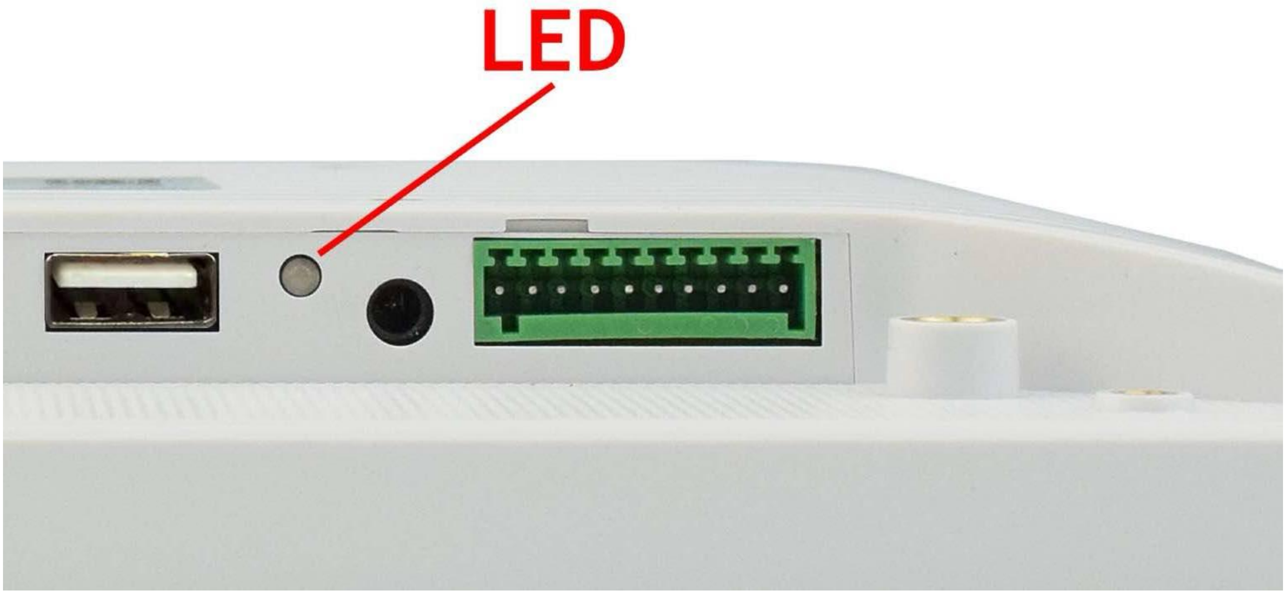
Microphone

The product has a front microphone for audio input.

Buttons and Status LED

Status LED

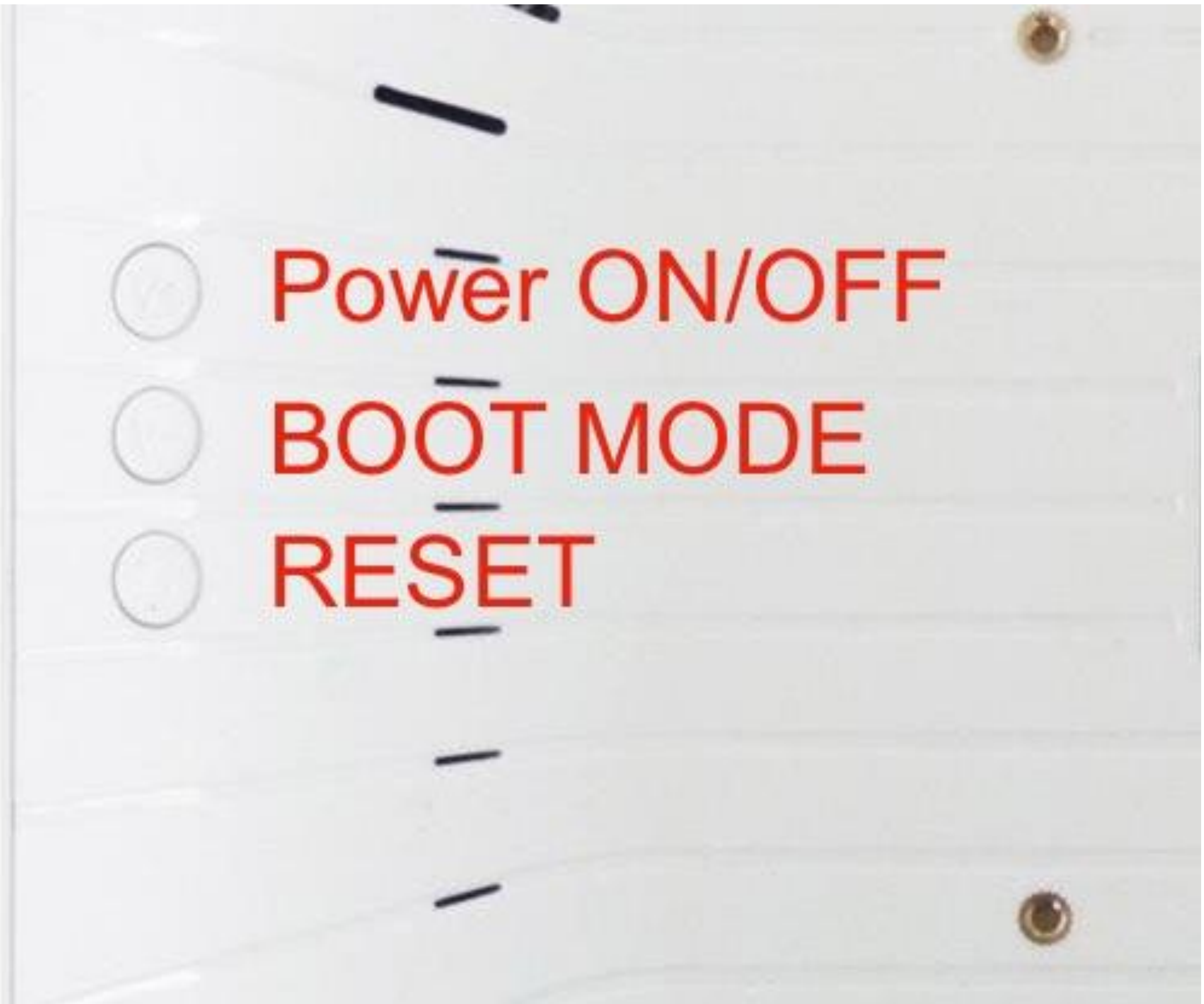
This product has a status LED. The LED turns RED after power on, turns GREEN when the system is booted.



Status LED

Buttons

There are 3 buttons on the back side: Power on/off, BOOT and RESET.



Buttons

What every button does:

- PWR:
    a. Before powering on: press to power on.
    b. After powering on: hold 5 seconds to force power off.
- BOOT Mode: Press the button before and while powering on: enter flashing mode to download program to the device.
- RESET: Reboots the device.

Mounting Procedure

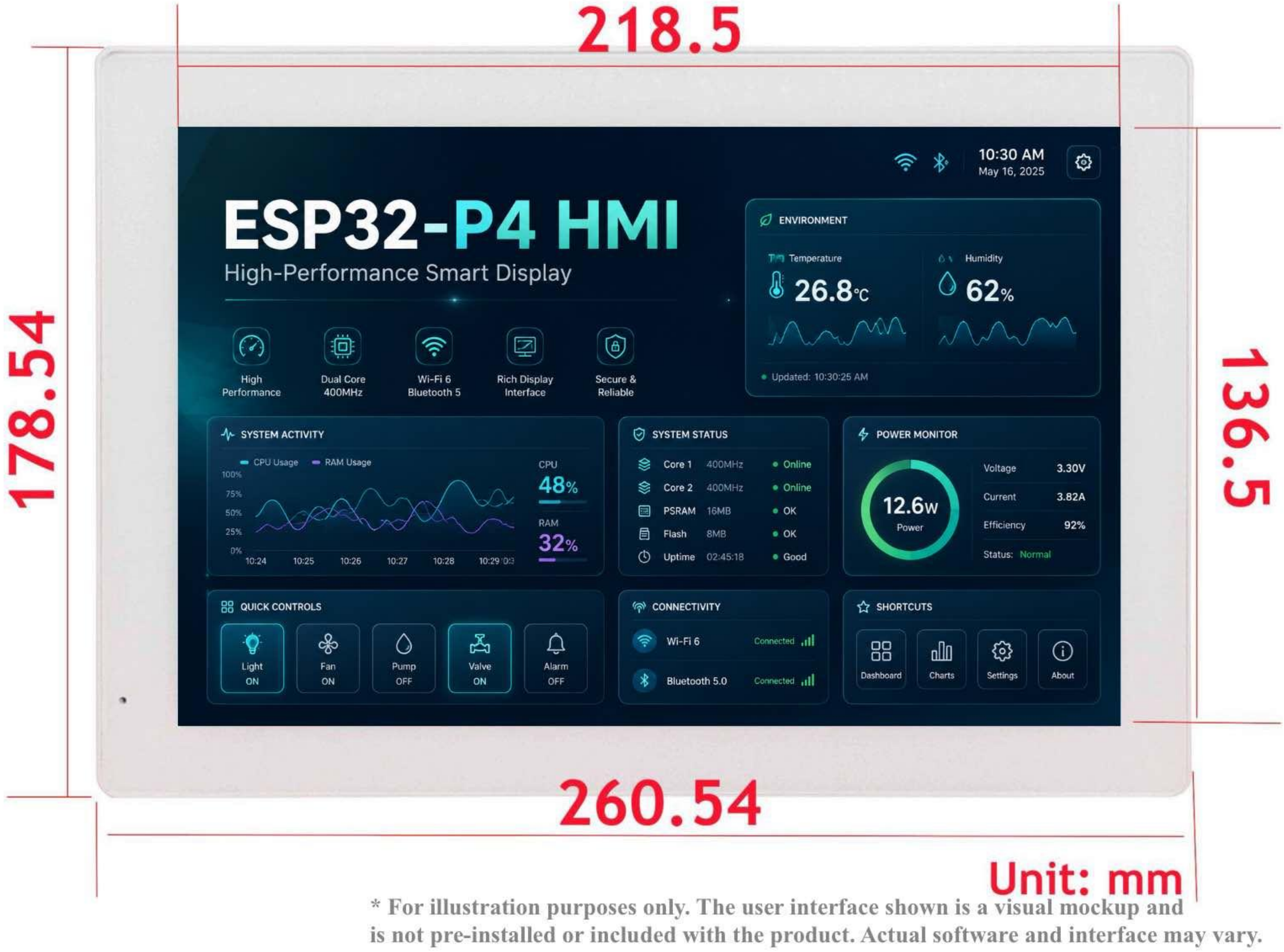
You can mount AIO-ESP32P4-101 with VESA mounting (guide): **75 x 75** mm, 4 x **M4** (6mm) screws.

The AIO-ESP32P4-101 can also be mounted on the supplied desktop stand. A base stand is included by default. If you do not require the stand, please specify this when placing your order.

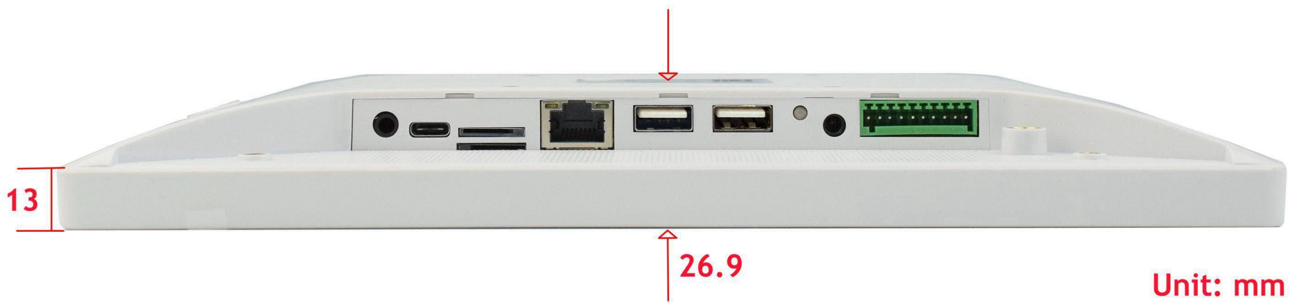
> **Attention**
>
> Do not apply excessive mechanical pressure to the display when mounting it in an enclosure.

Mechanical Specifications

The outer mechanical dimensions of AIO-ESP32P4-101 are 260.54 x 178.54 x 26.9mm (W × H × D). Please refer to the technical drawing in the figures below for details related to the specific product measurements.



Front-View Mechanical Drawing



Side-View Mechanical Drawing

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